

LESSON GUIDE

Technology Plus

Unit 3: Hardware

2.1.2 – IoT Devices

Lesson Overview:

In this lesson, students will explore IoT devices, their key features, and how they impact daily life.

Students will:

- Identify IoT devices and how they communicate.
- Explain how IoT devices impact daily life.
- Describe the functions and key features of common IoT devices.

Guiding Question: What are IoT devices, how do they communicate, and how do they provide convenience in daily life?

Suggested Grade Levels: 8-12

Technology Needed: No

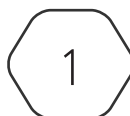
Approximate Time Required: 55-90 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 2.1 – Explain common computing devices and their purposes.
 - Internet of Things (IoT)
 - Home appliances
 - Home automation devices
 - Thermostats
 - Security systems
 - Home assistants
 - Deadbolts/door locks
 - Video doorbells
 - Vehicles
 - Internet Protocol (IP)/security cameras
 - Streaming media devices
 - Medical devices
 - Exercise equipment
 - Wearable devices

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Lesson Background

Background Information:

This lesson is a continuation of exploring common computing devices and their purposes with a focus on smart devices, also known as Internet of Things (IoT) devices. Smart is used to describe IoT devices on the activity cards. Students should understand that IoT devices are also computing devices that we interact with and utilize daily.

Students will explore IoT communication methods, common IoT devices, their key features, and how they can positively impact daily life. By the end of the lesson, students should be able to identify and describe common IoT devices and recognize how the devices impact their personal life.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 2.1.2 – IoT Devices • Activity Slides 2.1.2 – IoT What Am I
Materials per Group	• Activity 2.1.2 – IoT What Am I Cards (cut out)
Materials per Student	• Student Notes 2.1.2 – IoT Devices • Students Handout 2.1.2 – IoT Devices • Assessment 2.1.2 – IoT Devices (optional)

Term	Definition
Internet of Things (IoT)	a network of interconnected devices that communicate and exchange data over the internet
IoT device	a smart device that senses, connects, processes data, and communicates to improve decision-making or automate tasks

2.1.2 – IoT Devices

Guiding Question: What are IoT devices, how do they communicate, and how do they provide convenience in daily life?

Lesson Launch (5-10 minutes)

What Makes a Device Smart?

- Prompt students to think about smart devices in their home.
- Ask students to name some of the smart devices they have.
- As students name some of the devices write them on a surface where all the students can see them.
- **Ask:** What makes those devices smart?
- Allow a few students to answer.

Students may say, “The Internet of Things makes devices smart by enabling them to collect, transmit, and act on data.”

Internet of Things (IoT)

- When you add internet to a device in your home, you are joining it to something called the **Internet of Things (IoT)**.
- The IoT allows devices to communicate and exchange data with each other over a network.
- IoT devices offer convenience, efficiency, and security in daily life.
- **Say:** Today we will explore IoT devices, their key features, and how they impact daily life.

Smart by Design: IoT Devices (25-35 minutes)

- Distribute Student Handout 2.1.2 – IoT Devices and encourage students to fill in the answers during the lesson.

IoT Communication Methods

- Introduce IoT communication methods.
 - Long range covers a wider area with lower data rates, while short range covers less area with higher data rates.
 - Data rate refers to the amount of data transmitted during a specified period over a network.
 - Wired communication range is determined by the cable length.
 - All communication types are not listed, only common types.
 - Short Range Wireless Communication Types:
 - Wi-Fi: high data rate, suitable for smart home/office devices.
 - Bluetooth or BLE: medium data rate, suitable for wearable and short-range sensors.

- Near Field Communication: low data rate, suitable for mobile payments or smart access controls.
- Long Range Wireless Communication Types:
 - Cellular: high data rate, suitable for smart vehicles.
 - LoRaWAN: low data rate, suitable for smart agriculture.
 - Smart agriculture refers to IoT devices used on farms to increase the quality and quantity of crops while optimizing human labor. i.e. knowing when soil is too dry or too wet because of a sensor.
- Wired Communication Types:
 - Ethernet: high data rate, suitable for industrial settings.
 - Powerline Communication: medium data rate, suitable for smart grids.
 - A smart grid is an electrical power distribution system, it uses IoT devices to transmit electricity from a power plant to homes and businesses.

IoT Devices

- Introduce **IoT devices**, and discuss the common types, and their key features.
 - Mention that IoT devices are still computing devices.
 - Just because a device uses the connection methods previously discussed does not mean it is an IoT device.
 - An IoT device senses, connects, processes data, and communicates to improve decision-making or automate tasks.
 - IoT devices do not need internet to work, some devices will work by being plugged in, charged, or with batteries.
 - As you introduce and discuss the devices, refer to the devices students mentioned at the beginning and point out where they fit in.
 - Introduce home appliances and their key features.
 - Introduce home automation devices and their key features.
 - Common home automation devices are thermostats, security systems, home assistants, deadbolts/door locks, and video doorbells.
 - Thermostats:
 - Integration: Connect with other smart devices and systems for unified control.
 - Voice Control: Operate devices using voice commands through virtual assistants.
 - Scheduling: Automate routines such as lighting, temperature, and appliance usage.
 - Security Systems:

- Remote Monitoring: View live camera feeds and receive security alerts remotely.
- Automation: Integrate with other smart devices to automate security measures.
- Alerts and Notifications: Receive immediate notification of security breaches or unusual activity.
- Home Assistants:
 - Voice Commands: Control smart home devices, play music, and receive information using voice commands.
 - Integration: Work with other smart devices and services to create a connected ecosystem.
 - Personalization: Offer personalized responses and routines based on user preferences.
- Deadbolts/Door Locks:
 - Remote Access: Lock or unlock doors from anywhere using a smartphone app.
 - Keyless Entry: Use codes, biometric data, or smartphone credentials for entry.
 - Access Logs: Track who accessed the home and when.
- Video Doorbells
 - Live Video Feed: View real-time videos of visitors at the door via a smartphone app.
 - Two-Way Audio: Communicate with visitors using built-in microphones and speakers.
 - Motion Detection: Receive alerts when motion is detected near the door.
- Introduce IoT-enabled vehicles and their key features.
- Introduce internet protocol/security cameras and their key features.
- Introduce streaming media devices and their key features.
- Introduce IoT-enabled medical devices and their key features.
- Introduce IoT-enabled exercise equipment and their key features.
- Introduce wearable devices and their key features.

Activity: IoT What Am I? (15-25 minutes)

- Divide students into groups of 3-4.
- Give each group a full set of IoT “What Am I?” cards (16 total).
- One student per round will hold a card on their forehead so only the group can see it.
- Rotate roles each round so everyone gets a turn.
- (Optional) Use a timer to keep rounds fast and fun.

Game Roles/Rules: Guessing Version

- **Student 1:** Holds the card on their forehead. Asks yes/no questions (“Am I used for home security?”).
- **Student 2:** Answers only “yes” or “no.”
- **Student 3:** Makes sure the rules are followed.
- **Student 4 (if present):** Joins as another answerer or monitor.
- **Important Rule:** Student 1 must say “Am I [device name]?” to count as correct.
- **Hint:** Encourage questions about key features of the device (e.g., “Do I connect with Wi-Fi?”).

Game Roles/Rules: Clue Version

- **Student 1:** Holds the card on their forehead.
- **Student 2:** Describes the device without using its name (“You sit it on a shelf and play music when talked to.”).
- **Student 3:** Monitors for rule-breaking (no blurting the answer, no peeking).
- **Student 4 (if present):** Joins as a describer or monitor.
- **Hint:** Focus on functions and benefits (“It can track your steps.” or “It can show you who’s at the door.”).

Putting It All Together (5-10 minutes)

- Review what students learned today by discussing the questions below as a class.
- What are IoT devices?

IoT devices are physical objects with sensors and software that connect to a network. They can collect, share, and act on data to improve decision-making or automate tasks.

- How do IoT devices communicate with one another?

IoT devices use short-range methods like Wi-Fi, Bluetooth, or NFC; long-range methods like Cellular or LoRaWAN; and wired methods like Ethernet or Powerline Communication.

- What are some common types of IoT devices?

Common IoT devices include smart home appliances, automation devices (thermostats, security systems, door locks), vehicles, cameras, streaming devices, medical monitors, exercise equipment, and wearables.

- How do IoT devices provide convenience in daily life?

IoT devices make life easier by adding automation, remote access, and integration. They save time, improve efficiency, and increase security – for example, adjusting a thermostat from your phone or getting alerts from a smart doorbell.